INCH POUND
MIL-PRF-1/28H
22 July 1997
SUPERSEDING
MIL-E-1/28G
3 December 1970

PERFORMANCE SPECIFICATION SHEET
ELECTRON TUBE, PHOTOMULTIPLIER
TYPE 1P21

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: 9-stage, side-on, S4 spectral response

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ebb	Eb-Edy9	Light Source	Light	Ib	ib	TA	Rp	Pp	Cathode area	Shock	Alt
Unit:	V dc	V dc	- - -	ℓm	mA dc	ma	°C	Ohms	W	Lighted sq cm	G	ft
Maximum:	1,250 (Note 20)	250 (Note 20)	- - -		0.1	1.0	75	Note 3	0.25	- - -	200	40,000
Test Conditions:	1,000 (Note 1)	100 (Note 2)	LS1	10^{-2}	- - -	- - -	- - -	10,000	- - -	0.8 (Note 4)	- - -	- - -

GENERAL:

Qualification - Required

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Table 1. Testing and Inspection.

Inspection	Method	Conditions	Symbol	Limits Min	Limits Max	Units
Qualification inspection						
Low-frequency vibration	1031	No voltages; t = 60 minutes	- - -	- - -	- - -	- - -
Direct-interelectrode capacitance	1331	See note 5	C	2.0	6.0	pF
Direct-interelectrode capacitance	1331	See note 6	C	3.5	9.5	pF
Low-pressure breakdown	- - -	Ebb = 1,250 V dc; Eb to Edy9 = 125 V dc; voltages for all other stages = 125 V dc per stage; Rp = 1 Meg (see note 7, 8 and 9)	- - -	- - -	- - -	- - -
Spectral response identification	5303	S4	- - -	- - -	- - -	- - -
Plastic base material	- - -	See note 13	- - -	- - -	- - -	- - -
Quality conformance inspection, part 1						
Cathode current	- - -	Light = 0.01 ℓm (see notes 10, 11, 12 and 19)	Ik	0.20	- - -	μA dc
Anode current	5301	See notes 10, 12 and 19	Ib	400	- - -	μA dc
Dark current	5301	Light = 0 (see note 19)	Lik	- - -	2.5	μA dc
Dark current (1)	5301	Light = 0 (see note 19)	Lib	- - -	0.05	μA dc
Dark current (2)	5301	Light = 0; Ebb = 1,250 V dc (see note 19)	Lib	- - -	1.0	μA dc
Dark current (3)	5301	Ebb = 500 V dc; Eb to Edy9 = 50 V dc; Rp = 10,000 ohms (see note 19)	Lib	- - -	0.006	μA dc
Dark current (4)	5301	Light = 0 (see notes 14 and 19)	Lib	- - -	0.01	μA dc
Signal to noise ratio	5306	Light = 10^{-8} ℓm; Rp = 0.1 Meg (see notes 15 and 19)	S/N	48	- - -	dB
Quality conformance inspection, part 2						
Secureness of base, cap or insert	1101	See note 16	- - -	- - -	- - -	- - -
Shock	1041	200G ± 10% (see note 17)	- - -	- - -	- - -	- - -
Base pin solder depth	1111		- - -	- - -	- - -	- - -
Permanence of marking	1105		- - -	- - -	- - -	- - -

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Table 1. Testing and Inspection -Continued.

Inspection	Method	Conditions	Symbol	Limits Min	Limits Max	Units
Quality conformance inspection, part 3						
Life-test provisions	- - -	Group C; t = 500 hours (see note 4)	- - -	- - -	- - -	- - -
Life-test end points	- - -	Anode current (see notes 10 and 18)	- - -	- - -	- - -	- - -

NOTES:

1/ Overall anode to cathode dc supply voltage.

2/ Anode to 9th dynode dc voltage.

3/ An anode load resistance of at least 10,000 ohms is recommended for a protective resistance.

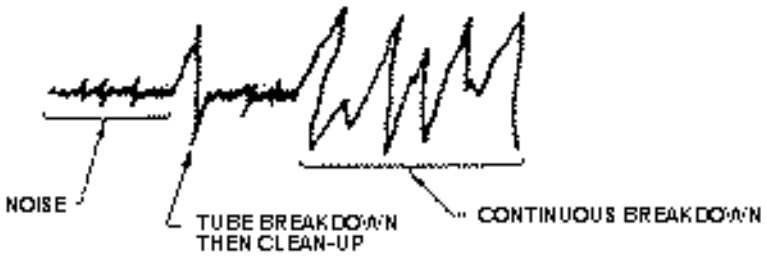
4/ Ebb = 1,000 V adjust luminous flux at cathode initially to obtain Ib = 0.1 mA dc. Color temperature shall be greater than 2,000 °K.

5/ Capacitance between anode and dy9, all other elements grounded.

6/ Capacitance between anode and all other elements.

7/ Test circuit and test equipment requirements:

A bleeder network to provide equal voltage per stage; 8-megohm series resistance for cathode, and all dynodes excluding the last dynode (9th); 1-megohm series resistance for last dynode (9th) and anode; a regulated (0.1 percent) power supply with low ripple (0.05 volts peck-to-peck maximum) as a voltage source; a sensitive dc microammeter to determine anode current; an oscilloscope, with a low-capacity divider, is connected across the anode load resistance (1 megohm) to monitor noise and breakdown; base pins of photomultiplier under test connected so that the degradation of breakdown characteristics is minimized; a vacuum pump, low-pressure chamber and manometer; a light source with an adjustable iris and sandblasted glass. Mask tube with lightproof material. The light source should be located so that light from it is striking the cathode of the photomultiplier. (Light from the light source should be the only light striking the cathode.) The opening of the iris is varied to obtain approximately 10 microamperes of anode current. Observe the tube noise signal as displayed by the oscilloscope. Reduce pressure of vacuum chamber to 130 mmHg. The tube should be subjected to this pressure for a period of 2 minutes. During pressure reduction, the tube noise signal, as displayed by the oscilloscope, should be observed to determine tube breakdown. When breakdown occurs, the signal being monitored by the oscilloscope will resemble the following:



When continuous breakdown occurs, a change in anode current will result.

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NOTES:

8/ Vary light to obtain approximately 10 microamperes of anode current.

9/ The tube shall operate without indications of breakdown in an evacuated chamber in which the pressure does not exceed 130 mmHg absolute.

10/ Adjust tube position to give maximum sensitivity.

11 Apply 100 volts dc between cathode and all other electrodes. The pins shall be tied together.

12/ The light during test, except during life test, shall be incident upon an aperture approximately 0.8 inch by 0.2 inch, located to center on the cathode.

13/ With the qualification test report, the tube manufacturer shall certify that the base used has been fabricated from an insulating material qualified under MIL-M14/H for type CFG or CGF. Bases previously qualified to method 1216 are acceptable.

14/ The overall anode to cathode voltage in this test shall be determined as follows:

With a light flux of 10^{-5} lumens, the supply voltage, epp, shall be varied until the anode current is 180 μA. With light flux set at 0, this voltage now serves as the test voltage.

15/ Prior to testing, the tubes shall be kept in the dark at an ambient temperature of 70° ± 5° F for not less than 1 hour. The light spot on the photocathode, based on a projected dimension perpendicular to beam axis, shall be 0.125 to .188 inch (3.18 to 4.76 mm).

16/ Immersion in water shall be omitted and tubes shall be tested dry. Torque shall be applied with 20.0 in-lb minimum.

17/ After the shock test is performed, the tube shall meet the requirements of all the test designated as quality conformance inspection tests, part 1.

18/ Anode current limit is 40 μA dc minimum or 1/20 of initial Ib, whichever occurs first.

19/ This test to be performed at the conclusion of the holding period.

20/ An ac supply voltage may be used having a peak value equal to this maximum value.

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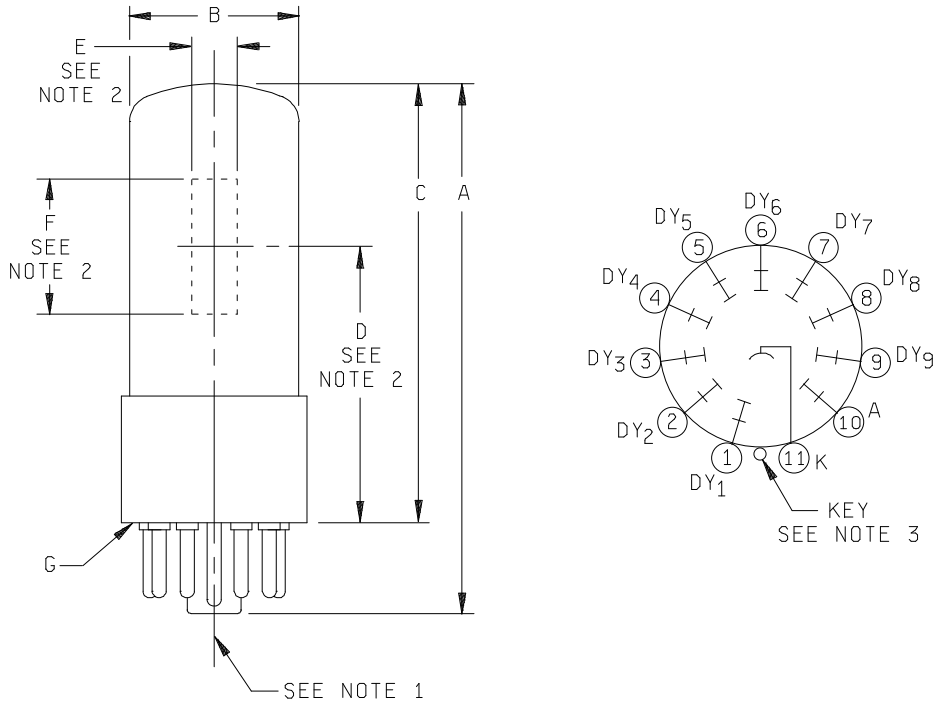

Dimensions in inches with metric equivalents (mm) in parentheses		
Ltr	Minimum	Maximum
Qualification inspection		
C		3.125 (79.38)
E	.312 (7.92)	
F	.937 (23.80)	
G	Base: B11-88 (see note 3)	
Quality conformance inspection, part 2		
A		3.687 (93.65)
B		1.187 (30.15)
D	1.843 (46.81)	2.031 (51.59)

NOTES:

1/ Centerline of bulb shall not deviate more than 2° in any direction from the perpendicular erected at center of bottom of base.
2/ Dimensions D, E and F refer to the usable cathode area.
3/ Photosensitive surface of cathode shall be oriented with respect to base guide key.

FIGURE 1. Outline drawing of electron tube type 1P21.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - AS, CG, MC, OS
Air Force - 11, 19, 99

Preparing activity:
DLA - CC

(Project 5960-3437)

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